

CSE4063 Fundamentals of Data Mining Project#1

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Problem Definition

This project aims to predict recipe star ratings (1-5 scale) using user interaction data and review characteristics from the Recipe Reviews and User Feedback Dataset. The goal is to develop classification models that can accurately predict user ratings based on features such as user reputation, comment engagement metrics (thumbs up/down, reply count), and other available attributes. This prediction task will help understand what factors influence recipe ratings and could be valuable for recipe recommendation systems.

Dataset

The data was collected from the Taste of Home cooking recipe website for sentiment analysis of food reviews by researchers from Warsaw University of Technology.

The dataset was compiled to include the top 100 most popular recipes, chosen for their popularity among users. It includes 18182 instances and 15 features.

The researchers used the Selenium library to systematically extract data from the HTML pages of each recipe.

Dataset



Recipe Reviews and User Feedback

Donated on 10/23/2023

The "Recipe Reviews and User Feedback Dataset" is a comprehensive repository of data encompassing various aspects of recipe reviews and user interactions. It includes essential information such as the recipe name, its ranking on the top 100 recipes list, a unique recipe code, and user details like user ID, user name, and an internal user reputation score. Each review comment is uniquely identified with a comment ID and comes with additional attributes, including the creation timestamp, reply count, and the number of up-votes and down-votes received. Users' sentiment towards recipes is quantified on a 1 to 5 star rating scale, with a score of 0 denoting an absence of rating. This dataset is a valuable resource for researchers and data scientists, facilitating endeavors in sentiment analysis, user behavior analysis, recipe recommendation systems, and more. It offers a window into the dynamics of recipe reviews and user feedback within the culinary website domain.



Dataset Characteristics

Tabular, Other

Subject Area

Computer Science

Associated Tasks

Classification, Other

Feature Type

Real, Categorical, Integer

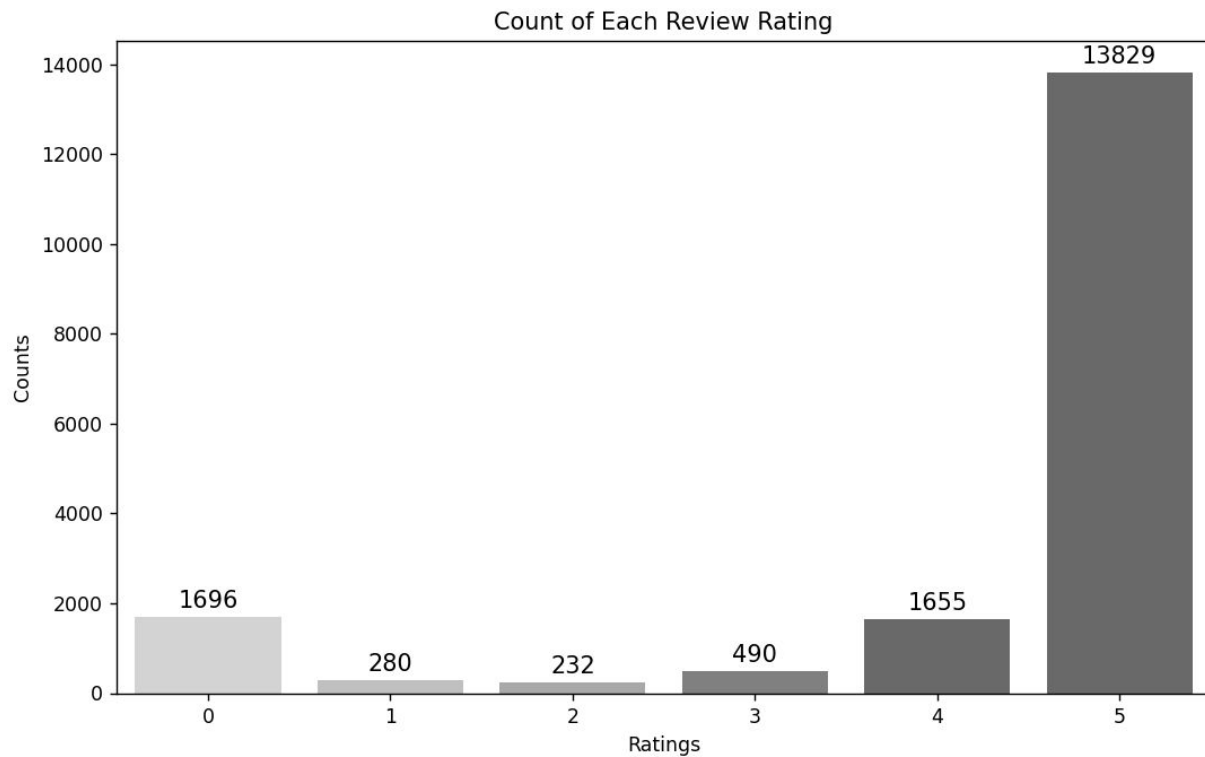
Instances

18182

Features

15

Dataset



Dataset

Features of the Dataset

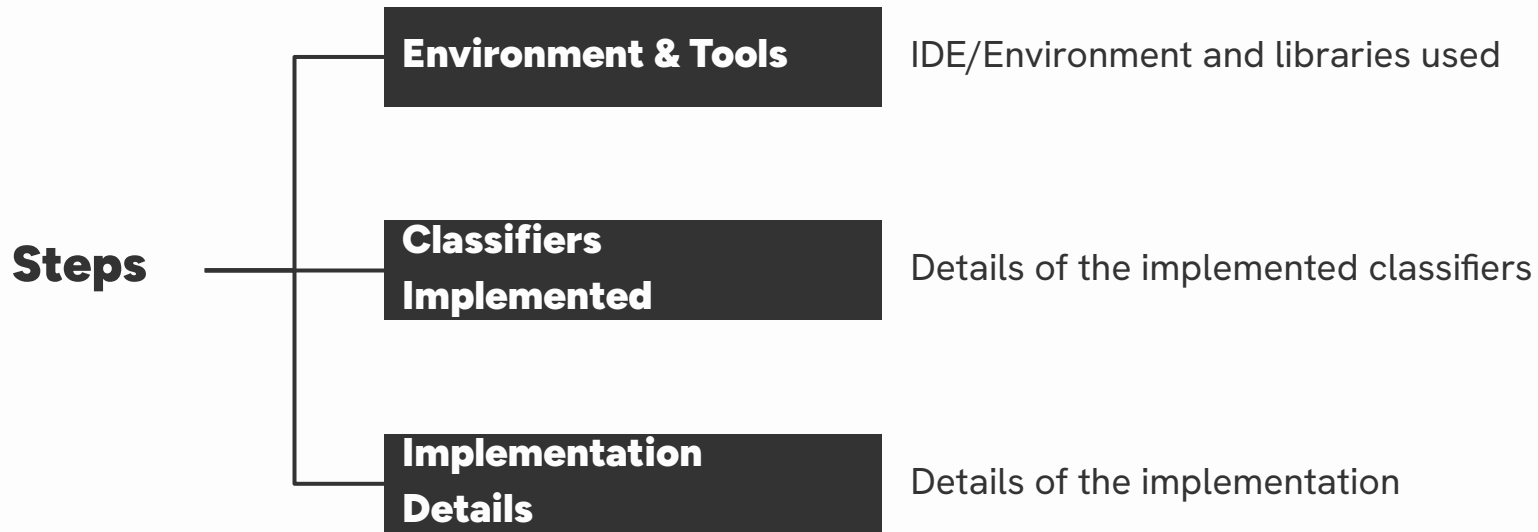
1. **Num records:** Number of records
2. **Recipe name:** Name of the recipe the comment was posted on
3. **Recipe number:** Placement of the recipe on the top 100 recipes list
4. **Recipe code:** Unique id of the recipe used by the site
5. **Comment ID:** Unique id of the comment
6. **User ID:** Unique id of the user who left the comment
7. **User name:** Name of the user
8. **User reputation:** Internal score of the site, roughly quantifying the past behavior of the user
9. **Create at:** Time at which the comment was posted as a Unix timestamp
10. **Reply count:** Number of replies to the comment
11. **Thumbs up:** Number of upvotes the comment has received
12. **Thumbs down:** Number of downvotes the comment has received
13. **Stars:** The score on a 1 to 5 scale that the user gave to the recipe. A score of 0 means that no score was given
14. **Best score:** Score of the comment, likely used by the site the help determine the order in the comments that appear in
15. **Text:** The text content of the comment

Data Preprocessing

Key Steps

- **Missing Values:** Filled missing values in the text feature with an empty string and removed rows without star value (0 star).
- **TF-IDF Transformation:** Applied *TfidfVectorizer* to convert text data into numerical features (max 1000).
- **Scaling:** Used *MinMaxScaler* for numeric features (user_reputation, reply_count, thumbs_up, thumbs_down, best_score).
- **Combining Features:** Combined TF-IDF and scaled numeric features into a sparse matrix using `scipy.hstack`.

Implementation



Implementation

Environment & Tools

- **IDE/Environment:** Code developed in Python using VSCode.
- **Libraries Used:**
 - **Data handling:** pandas, numpy.
 - **Preprocessing:** scikit-learn for TF-IDF, scaling, and splitting.
 - **Model Training:** scikit-learn classifiers including DecisionTreeClassifier, BernoulliNB, ComplementNB, LinearSVC, MLPClassifier, and ensemble methods.
 - **Sparse Data:** scipy for handling sparse matrices.
 - **Visualization:** matplotlib, seaborn for performance graphs and confusion matrices.

Implementation

Classifiers Implemented

- **Decision Trees:**
 - **Algorithms:** Gini Index and Entropy.
- **Naive Bayes:**
 - **Variants:** BernoulliNB and ComplementNB.
- **SVM:**
 - **Algorithm:** LinearSVC
- **Neural Networks:**
 - **Architectures:** 1 hidden layer (100 units) and 2 hidden layers (100, 50 units).
- **Ensemble Models:**
 - **Bagging:** Used simple gini decision tree as base estimator.
 - **Boosting:** AdaBoost with SAMME algorithm.

Implementation

Implementation Details

- **Preprocessing Pipeline:**
 - Handles text processing, scaling, and combining features into sparse matrices.
 - Split dataset into training and testing subsets using 80-20% ratio.
- **Model Creation:**
 - Defined in models.py using a dictionary structure for flexible model creation.
 - Functions `create_models` and `create_ensemble_models` encapsulate model definitions.

Implementation

Implementation Details

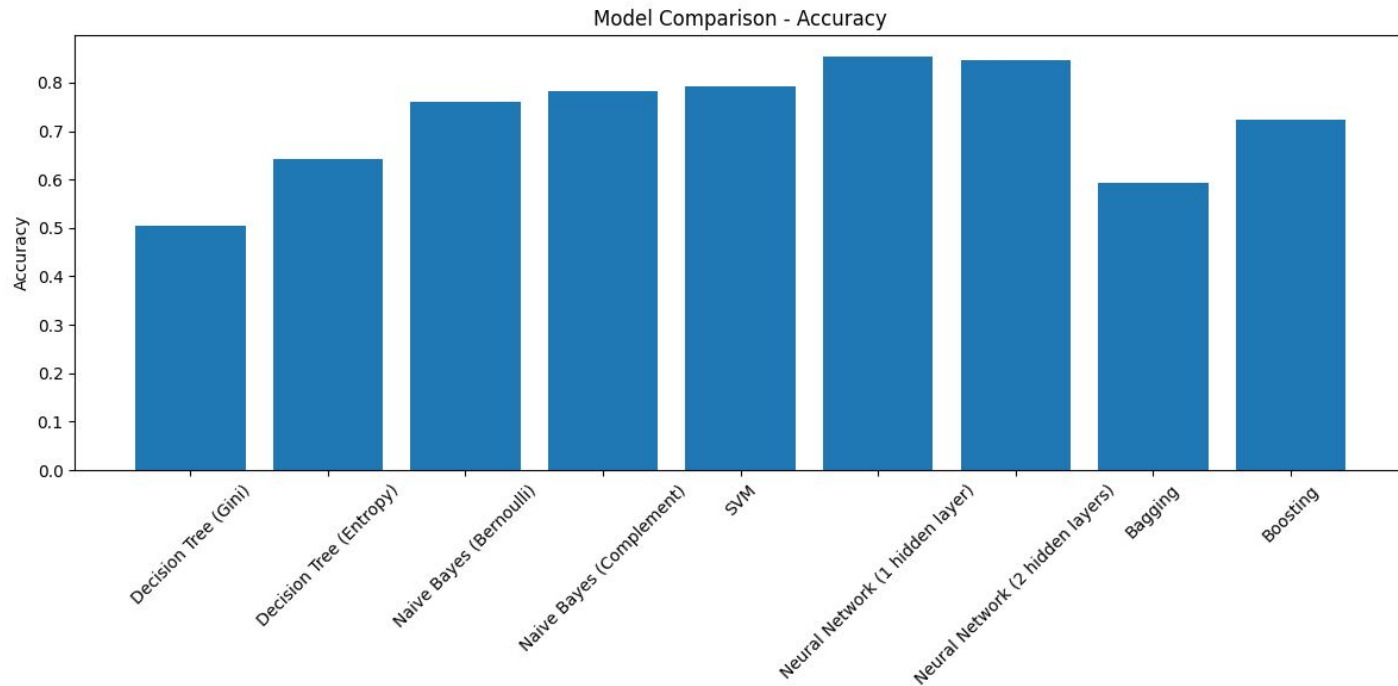
- **Training and Evaluation:**
 - Models are executed in `main.py`.
 - Logger class in `logger.py` saves runtime logs into text files for easy review.
 - Function `evaluate_model` in `evaluation.py` computes metrics (accuracy, precision, recall, F1-Score) and visualizes confusion matrices using `matplotlib` and `seaborn`.
 - Function `plot_comparison` visualizes metric comparisons across models.

Model Evaluation & Performance Results

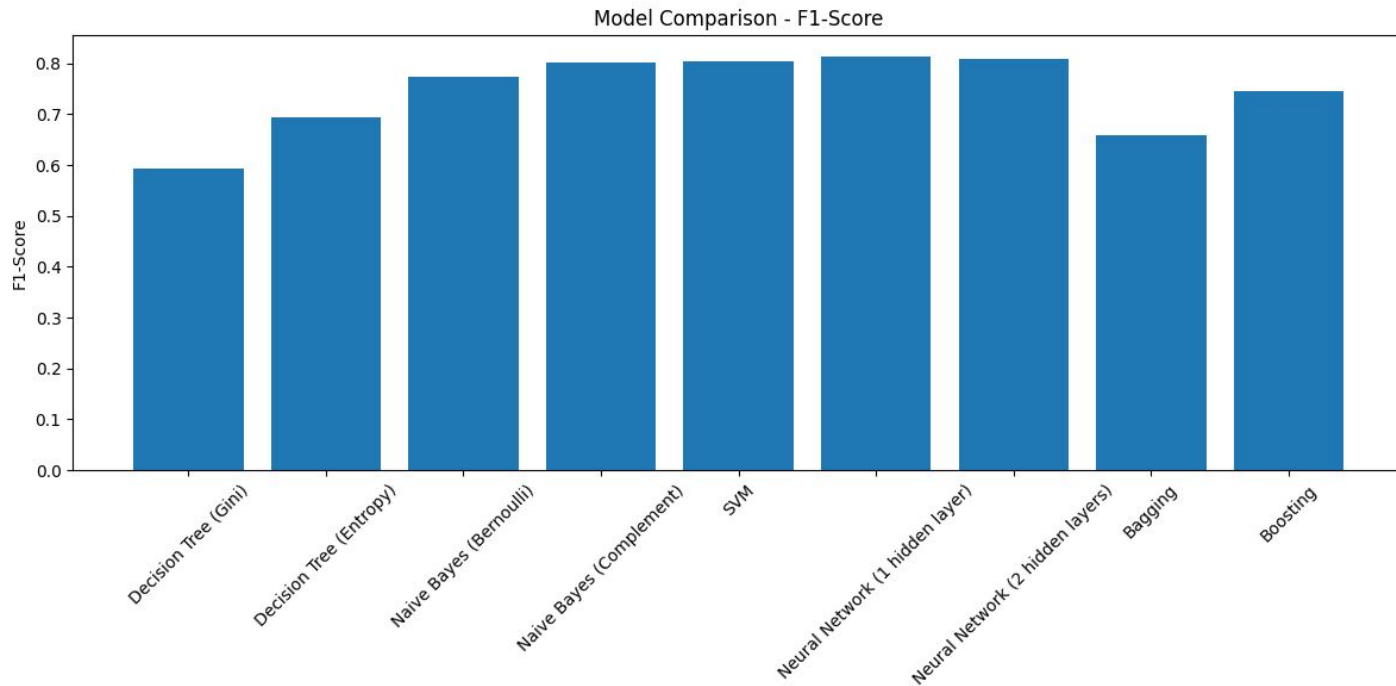
Models We Used

- Decision Trees (Gini and Entropy)
- Naive Bayes (Bernoulli and Complement)
- SVM
- Neural Networks
- Ensemble Methods (Bagging and Boosting)

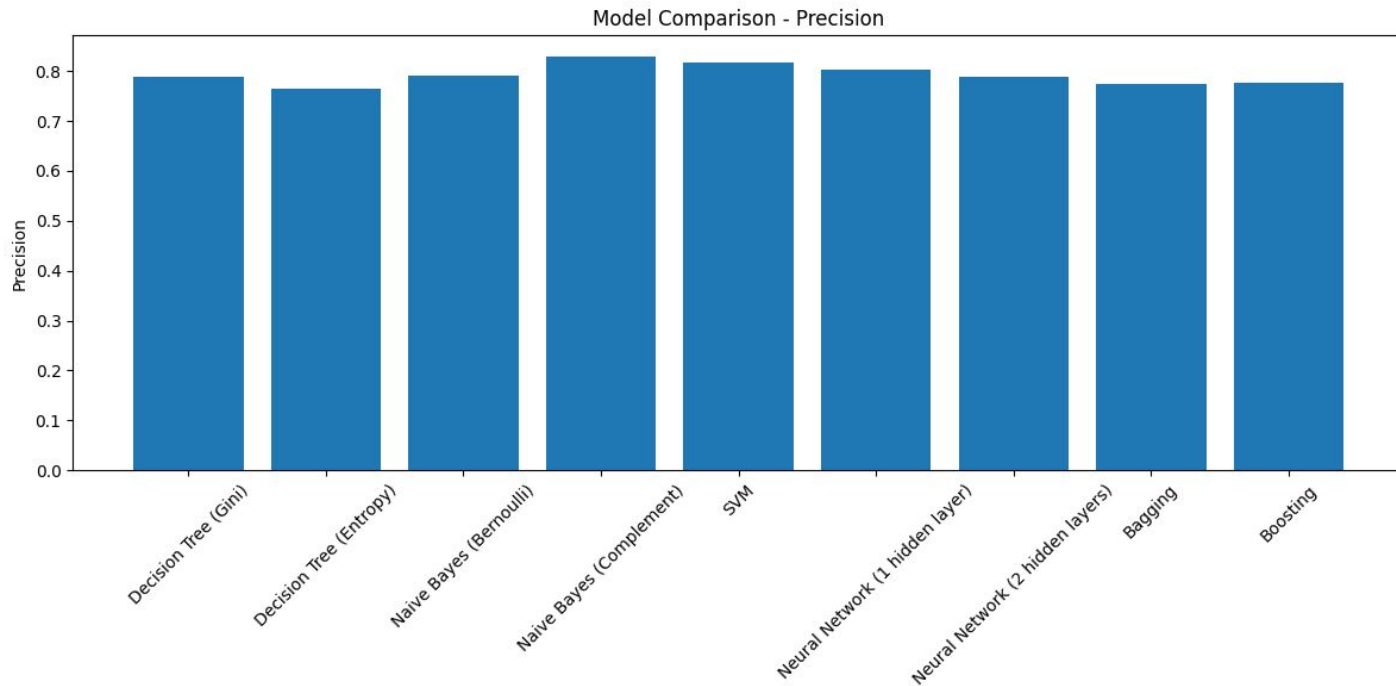
Model Comparison - Accuracy



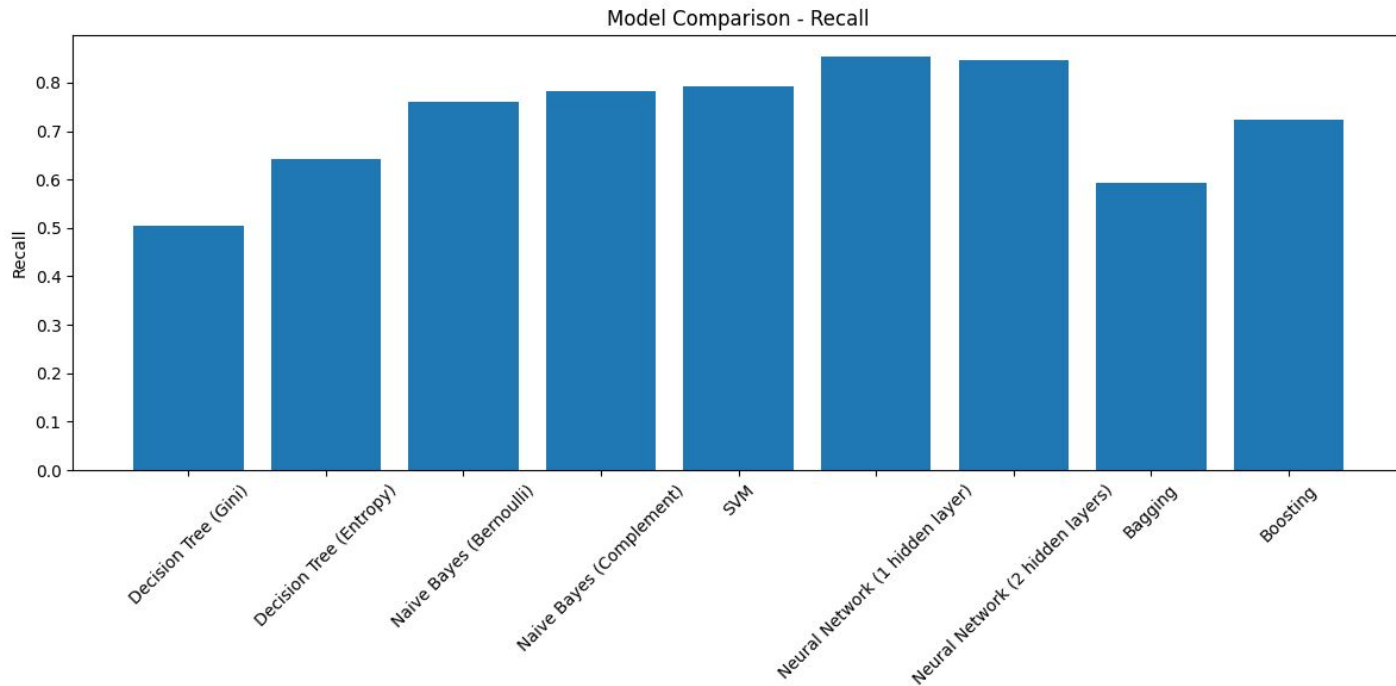
Model Comparison - F1 Score



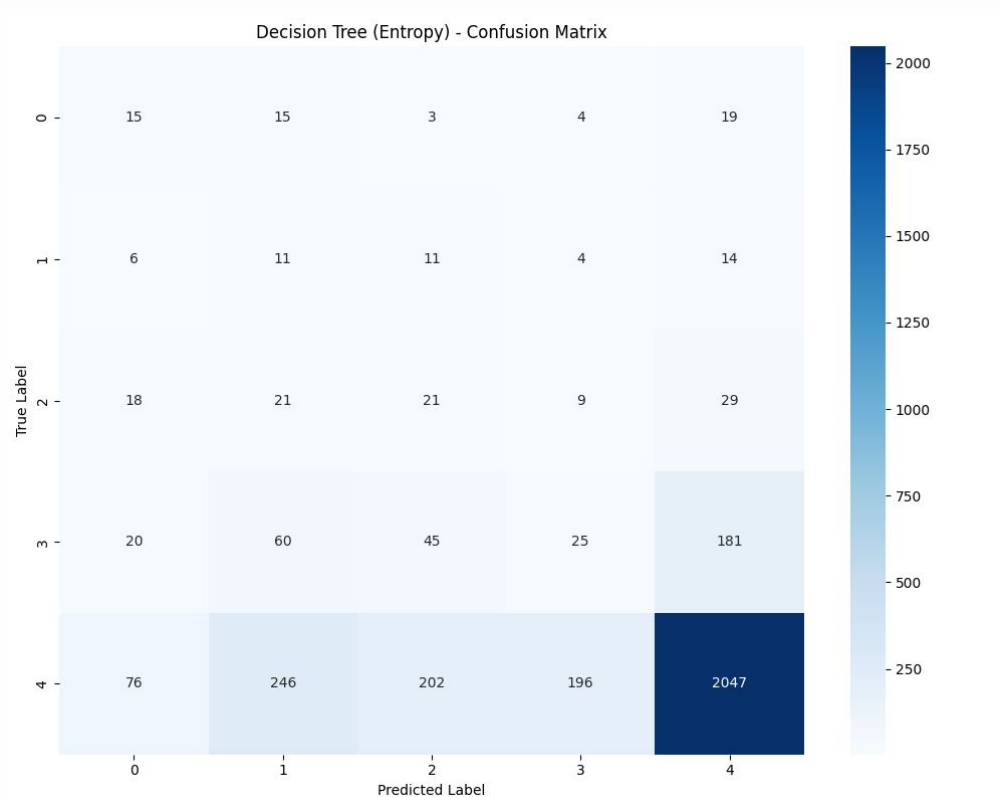
Model Comparison - Precision



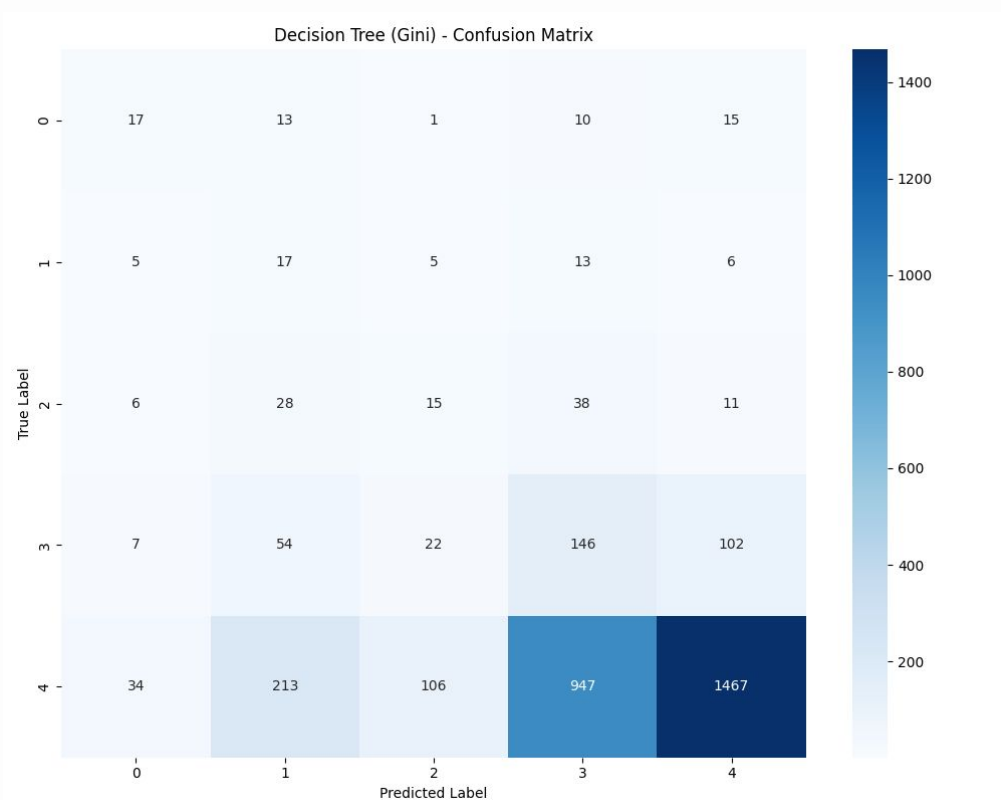
Model Comparison - Recall



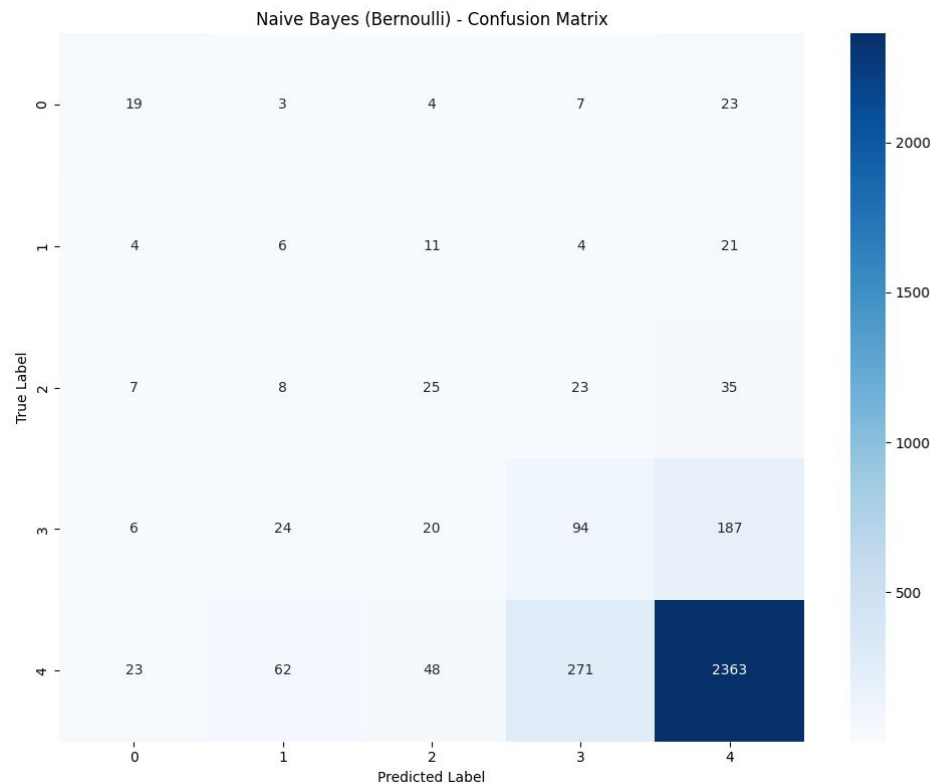
Decision Tree (Entropy) - Confusion Matrix



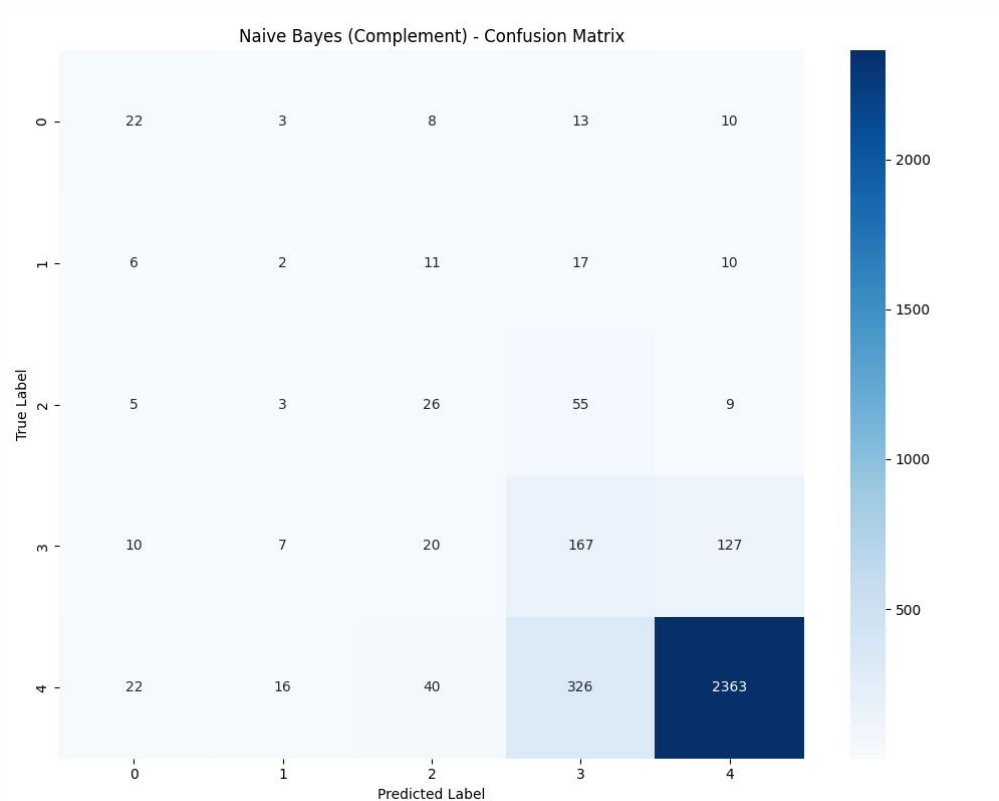
Decision Tree (Gini) - Confusion Matrix



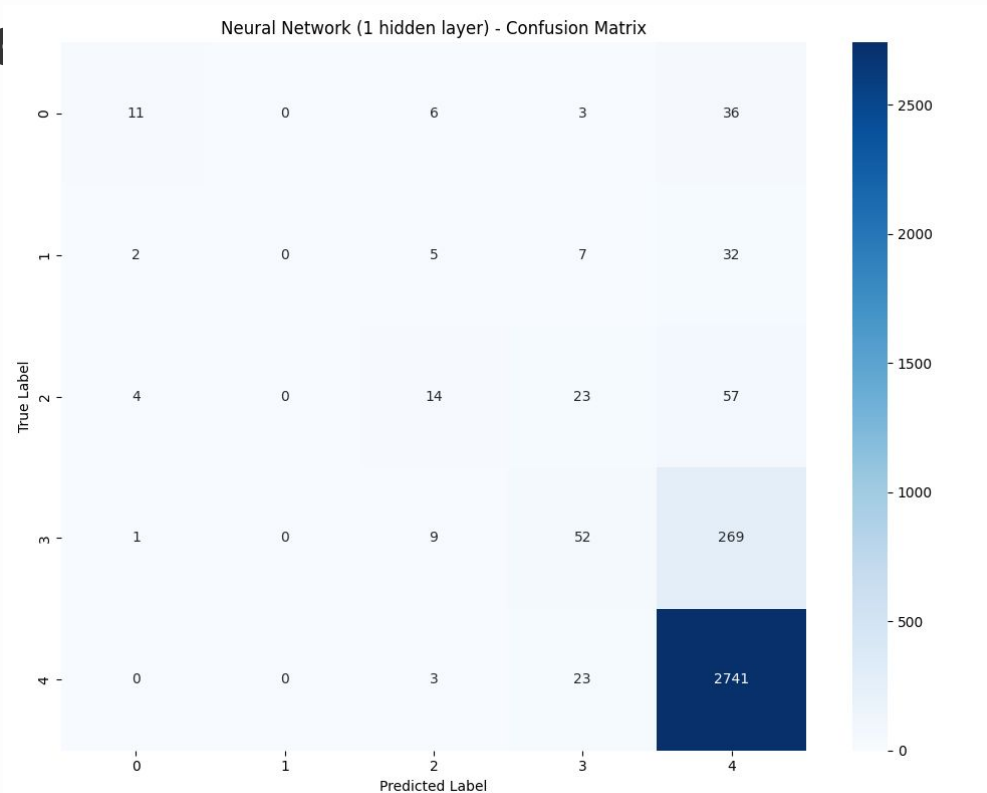
Naive Bayes (Bernoulli) - Confusion Matrix



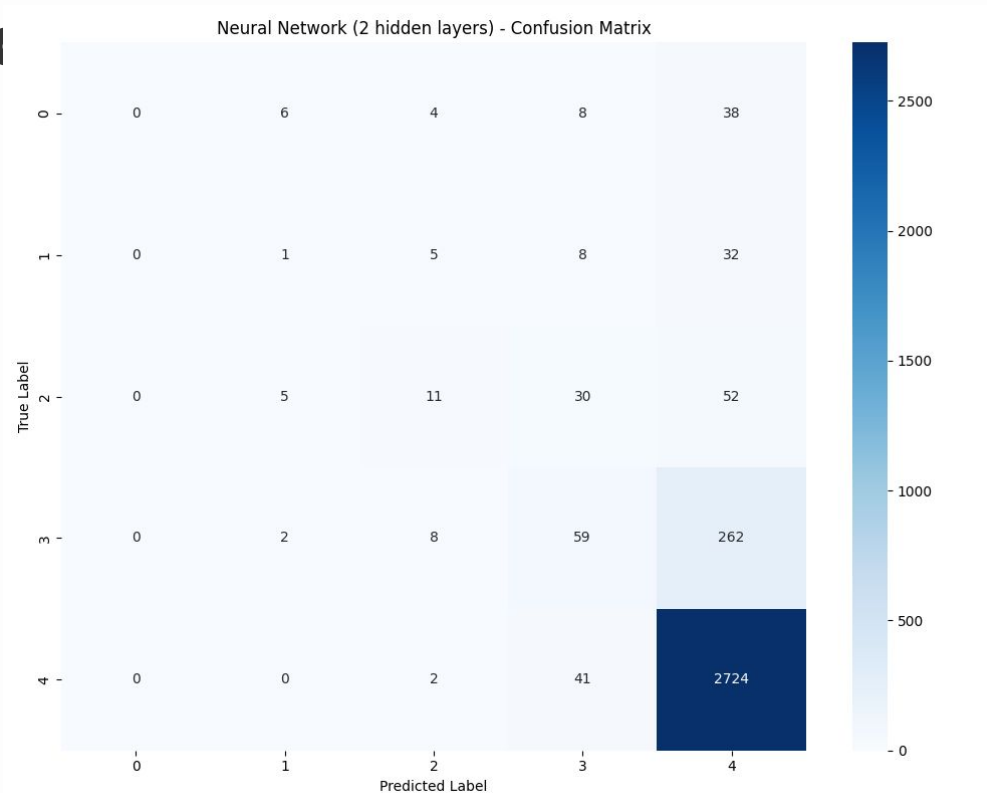
Naive Bayes (Complement) - Confusion Matrix



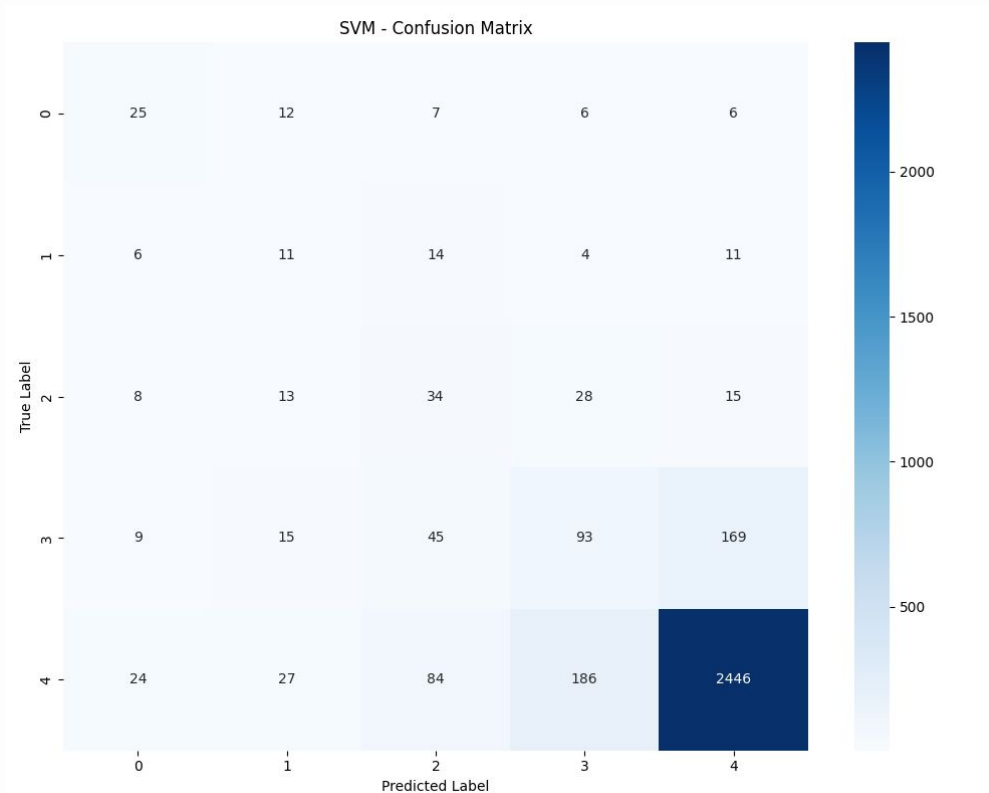
Neural Network (1 hidden layer) - Confusion



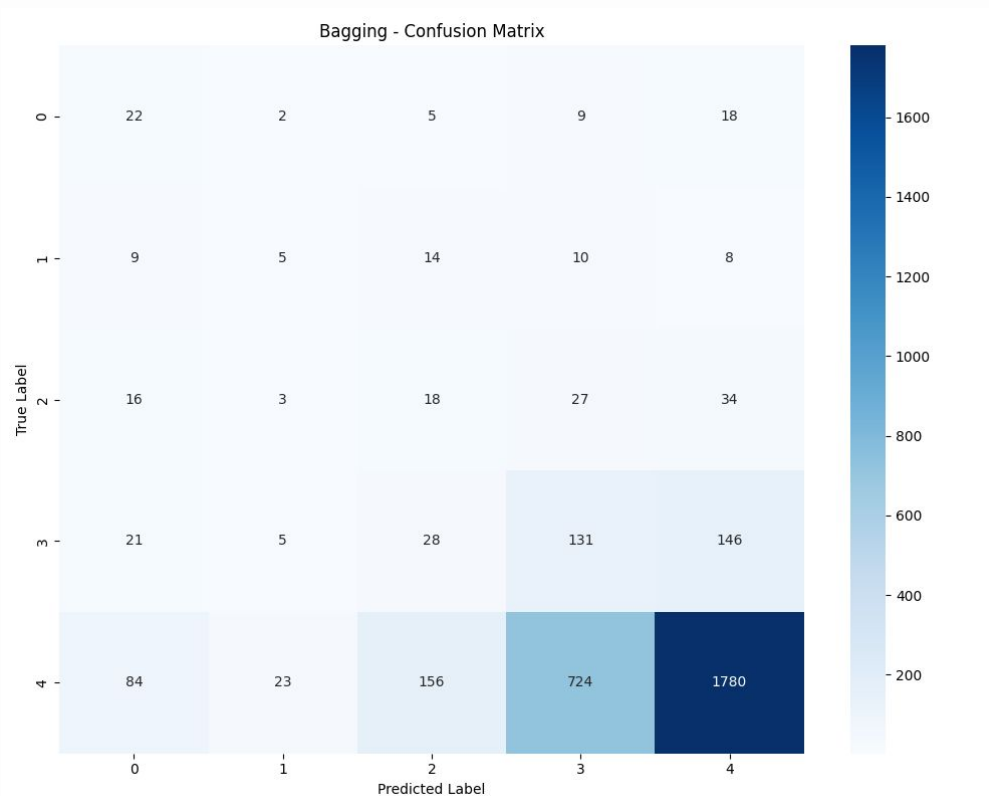
Neural Network (2 hidden layer) - Confusion



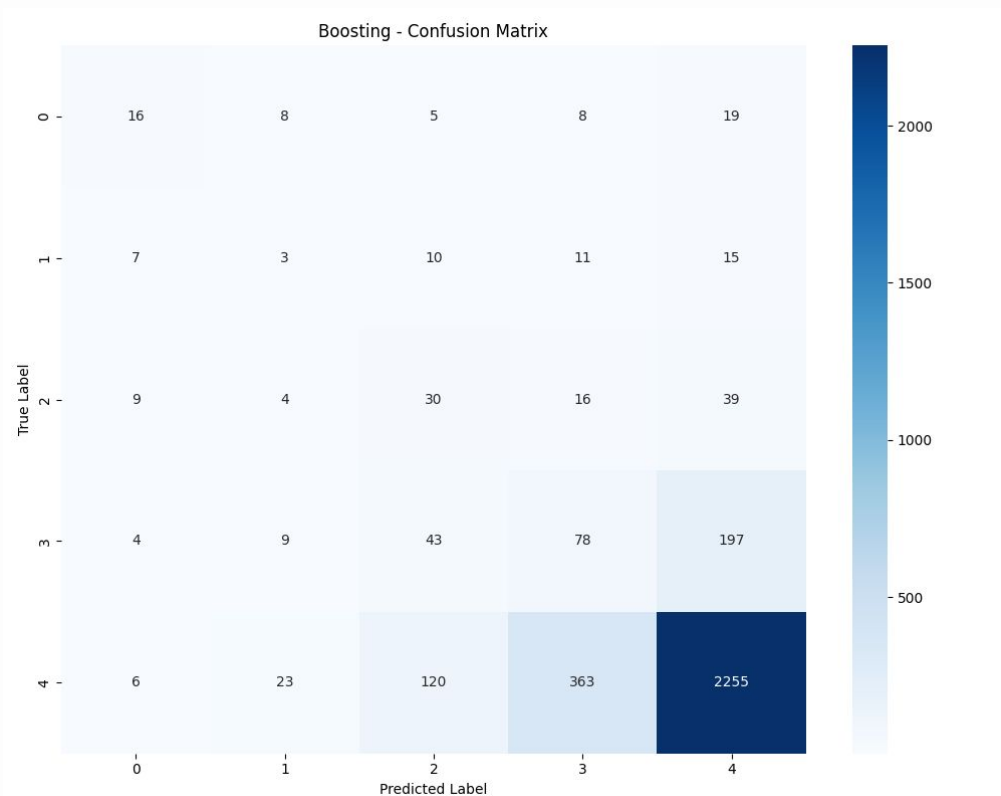
SVM - Confusion Matrix



Bagging - Confusion Matrix



Boosting - Confusion Matrix



Conclusions

Best Performing Models

- Neural Networks (Accuracy: 85%)
- SVM (Accuracy: 79%)
- Naive Bayes (Bernoulli and Complement) (Accuracy: 76-78%)

Model Behaviours

- High success in 5-star predictions
- Poor performance in middle ratings
- Class imbalance issues

References

Textual Taste Buds: A Profound Exploration of Emotion Identification in Food Recipes through BERT and AttBiRNN Models

By Amir Ali, Stanislaw Matuszewski, Jacek Czupyt, Usman Ahmad.

2023

Thanks!

Do you have any questions?